Zhan Jin

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Education

University of Illinois Urbana-Champaign, Champaign, IL *Aug 2026 - Present*
Ph.D. in Computer Science

- Advised by Prof. Francis Y. Yan

Peking University, Beijing, China *Sep 2022 - Jun 2026*
B.S. in Computer Science

- Outstanding Graduate of the Class of 2026 in the major of Computer Science and Technology, School of EECS

Research Interests

Building efficient, scalable systems that advance foundation model capabilities, and designing algorithms whose benefits grow as these capabilities improve.

Publications

Accelerating Cross-Model LLM Inference via Attention Map Reuse
Zhan Jin, Yuze Luo, Bin Cui, Xupeng Miao.
SIGMOD 2027

Tactic: Adaptive Sparse Attention with Clustering and Distribution Fitting for Long-Context LLMs
Kan Zhu, Tian Tang, Qinyu Xu, **Zhan Jin**, Yile Gu, Zhichen Zeng, Rohan Kadekodi, Liangyu Zhao, Ang Li, Arvind Krishnamurthy, Baris Kasikci.
ICLR 2026

Preprints

DualTune: Decoupled Fine-tuning for On-Device Agentic Systems
Rohan Kadekodi*, **Zhan Jin***, Keisuke Kamahori, Yile Gu, Sean Khatiri, Noah H. Bayindirli, Sergey Gorbunov, Baris Kasikci.
*Equal contribution

Research Experience

Data and Intelligence Research (DAIR) Lab, PKU *Sep 2025 - Apr 2026*
Advisors: Prof. Bin Cui and Prof. Xupeng Miao
Project: Accelerating Cross-Model LLM Inference via Attention Map Reuse

- Designed and evaluated cross-model attention-map reuse, enabling same-tokenizer models of different sizes to reach near-lossless accuracy across multiple benchmarks.
- Developed a runtime with optimized kernels and compressed attention maps, reducing TTFT by 15–20% and increasing throughput by up to 25% over full attention without reuse.

Systems for Future Intelligence (SyFI) Lab, UW *Jun 2025 - Sep 2025*
Advisor: Prof. Baris Kasikci
Project: Decoupled Fine-tuning for On-Device Agentic Systems

- Analyzed the performance and accuracy constraints in current on-device, tool-calling agent frameworks.
- Introduced a modular approach that separates tool selection and argument generation, identifying tool selection as the dominant error source.
- Proposed decoupled fine-tuning with specialized adapters for each subtask, improving tool-calling accuracy by 46% on MCP-Bench compared to the base 7B model.

Awards

- Outstanding Graduate of the Class of 2026 in the major of Computer Science and Technology, School of EECS
- Outstanding Research Award, Peking University, 2025
- Academic Excellence Award, Peking University, 2024
- Third Prize, Peking University Scholarship, 2024
- Second Prize, Integrated Circuit EDA Design Elite Challenge, Chinese Institute of Electronics, 2023

Skills and Interests

- Extensive programming experience with strong proficiency in C, C++, CUDA, and Python.
- Experienced in deploying and optimizing distributed training frameworks (Megatron-LM, DeepSpeed) and inference engines (vLLM, SGLang).
- Interested in Linux-based software development and embedded systems.